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Econometric analysis of pricing and energy policy regulations in Japan electric power exchange spot market

Mohamed Saad Suliman [a], Hooman Farzaneh [a,b,*]

[a] Interdisciplinary Graduate School of Engineering Sciences, Kyushu University, Fukuoka, Japan
[b] Transdisciplinary Research and Education Center for Green Technologies, Kyushu University, Fukuoka, Japan

A B S T R A C T

This study aims to provide a quantitative overview of the energy policy regulations concerning the Japan spot market liberalization and analyzes the market clearing price (MCP) and the regional spot prices based on the hourly contribution of the supply stack mix and the merit order effect. In addition, it estimates the regional spot prices interdependency on the MCP. The study acquires an hourly dataset in panel data format from 2016 to 2020 with total observations of 438,240 data records. The developed econometric models include panel data regression using fixed-effect estimator, ordinary least square, and two-stage least square equations. From the results, a one GWh addition from nuclear, hydroelectric, geothermal, biomass, solar, or wind power production technologies decreases spot prices. Contrarily, adding one GWh from thermal, hydroelectric storage, or regional interconnections raises the spot prices by 27.4, 606, 103 ¥/MWh, respectively. The spot prices of Hokuriku and Chugoku tend to offset the price spikes in Kansai, while Tokyo and Tohoku spot prices are significantly related. Adding one ¥/kWh to Tohoku spot price is estimated to raise the MCP by 0.35 ¥/kWh, while the addition of one ¥/kWh to Tokyo spot price is estimated to increase the MCP by 0.04 ¥/kWh. Furthermore, Kyushu and Tohoku, followed by Chubu, have appreciable control over the MCP. The study further highlighted policy implications to constrain the applications of hydroelectric storage as the costliest solution to balance the intermittency of renewable energy sources.

1. Introduction

1.1. Background

The advancement achieved in clean energy production technologies is expected to replace conventional fossil fuel energy in the future. The transition toward clean energy faces a rapidly growing energy demand along with the global warming indices. The deregulation of the electricity market has been implemented as an effective measure that addresses the energy supply and demand transactions in a commodity market platform. Besides the diversification of the energy resources that efficiently translated in the merit-order curve of the deregulated energy markets, the partially deregulated energy markets are correlated with the developing economies that continue to expand, build state capacity, and move toward democratic political institutions, which is reflected in the private electric utilities' competition for profits maximizations (Urpelainen and Yang 2019). On the other hand, consumers have the autonomy to select their commodity provider based on requirements. The electricity market liberalization is a long- and complex-term process requiring several market reforms. The complete shift from the vertically integrated oligopoly structure to the liberalized market consists of at least five major key elements (Williams and Ghanadan, 2006). First, commercialization of the state-owned companies to partial independent entities, that account for their financial records. Second, full privatization transfers the state-owned assets to the private sector. Third, unbundling, which is the pivotal part of the market reform to separate the various activities of the vertically integrated oligopoly structure. Fourth is the third-party access regulations to enable the authorized consumers to purchase energy directly from suppliers. Last is the competition through wholesale and retail platforms. The study conducted by (Grubb and Newbery, 2018) has summarized the evolution of the UK electricity market reform, where the five key elements were applied. Furthermore, the reform activities of the Chinese electricity market have constructed the retail competition (Jin et al., 2020).

1.2. Liberalization and energy policies evolution in the Japanese electricity market

In Japan, the decree dissolving of the Japan Electric Generation and Transmission Company has transferred the state-ownership to nine private regional companies (enacted in 1950). The Okinawa electric power company acceded as the tenth regional electricity utility in 1972. This structure premises a vertically integrated regional monopoly that controls the electricity from generation to the end-user. However,

* Corresponding author. Interdisciplinary Graduate School of Engineering Sciences, Kyushu University, Fukuoka, Japan.
E-mail address: farzaneh.hooman.961@m.kyushu-u.ac.jp (H. Farzaneh).

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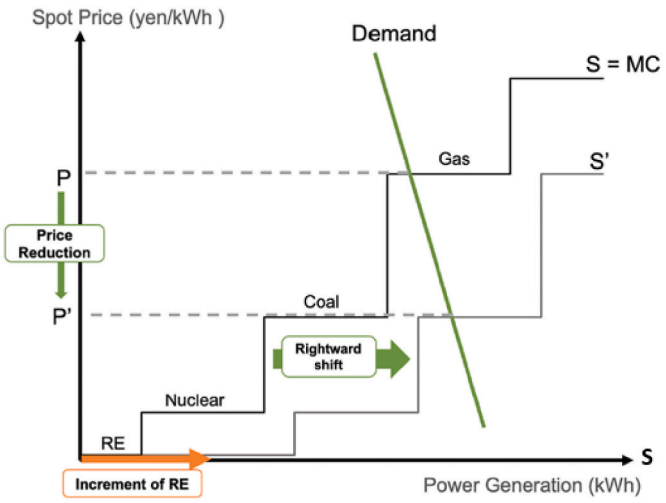


Fig. 1. Conceptual graph of the MOE (Sakaguchi and Fujii, 2021).

utilities have invested in aligning the growing energy demand with the rapid national economic growth. In 1995, the wholesale supply market was liberalized, followed by retail supply liberalization in 2000 for customers of extra high voltages (20 kV and above). By April 2005, the deregulation had been extended to involve all high voltage customers with a 50 kW and above contracted demand. Thereafter, the surplus solar power purchasing system was launched, which purchases power from the solar power plants with less capacity than 500 kW at a contract period of 10 years. Furthermore, the feed-in tariff (FIT) scheme, which sells back renewable energy directly to the grid, was implemented in July 2012. The Ministry of Economy, Trade, and Industry (METI) has recommended restraining the expansion of the liberalization scope that includes the household sector for further evaluation (Japan Electric Power Information Center (JPIC), 2019). In 2011, the consequences of the Great East Japan Earthquake and the resultant Fukushima Daiichi Accident were imposed to diversify the supply technologies to compensate for the nuclear power suspension, which is reflected in the electricity rates hike and supply-demand misbalance. Accordingly, METI has approved the policy from the report of the electricity system reform expert subcommittee to schedule the implementation of cross-regional coordination of transmission operators, unbundling the transmission and distribution sector, and empowering the whole retail market competition (METI, 2013). After the full liberalization of the electricity market in 2016, the market has 10 regional transmission network operators (TSO) with a limited scope of monitoring and controlling the regional networks encompassing Power Producers and Suppliers (PPS), retailing, and demand. The Organization for Cross-regional Coordination of Transmission Operators (OCCTO) is the independent system operator (ISO), which facilitates the nationwide grid operations, coordinates the cross-regional supply and demand balance, and performs unbiased and efficient technical management of the national grid. Although the last phase of (METI, 2013) policy is to activate the deregulated rates to the low voltage customers and legally unbundle the transmission and distribution grids by April 2020, the legal unbundling is implemented, but the regulated rates are maintained (JPIC, 2021).

The Japan Electric Power Exchange (JEPX) is the wholesale system operator, building the price mechanisms and facilitating spot and forward transactions. The JEPX trading volume contribution from the national demand was 1% until 2016, when market liberalization was extended to involve all the remaining sectors. The trading volume share is raised exponentially to reach 40% by the end of FY 2019 (JPIC, 2021). Due to the interconnection limitation, the market is composed of all the regional TSOs except Okinawa electric power company. The interconnections were managed on a first-come-first-served basis. However, from October 2018 onwards, the implicit auction approach is introduced, which restricts the use of interconnections to be through the spot market in day-ahead contracts (JPIC, 2021). The principal participants are electric power utilities and wholesale retailers, while special trading members are TSOs on behalf of FIT users and negawatt contracts of demand response aggregators.

1.3. Impact of renewable energy on shifting the merit order curve

The market operator synthesizes the merit order curve based on the demand and supply bids from the market participants. For each hour of the day ahead, the market operator arranges the supply bids in ascending price-quantity order, while the demand bids are ordered in descending price-quantity. The intersection of the aligned curves is referred to be the MCP and the market trading volume. Accordingly, various energy production technologies offer different Marginal Costs (MC) which are implied in hourly supply bids and, consequently the MCP (Farzaneh, 2019). Since the marginal cost of renewable energy is nearly zero, the higher contribution of renewable energy in the hourly bids, the lower MCP is detected, as depicted in Fig. 1. The phenomenon of shifting the MCP to a lower value while meeting the fixed demand at the typical trading volume is called Merit Order Effect (MOE) (Sensfuß et al., 2008).

The MOE is widely investigated by scholars to assess the impact of renewable energies on electricity spot prices. From the global electricity spot markets, the European Energy Exchange (EEX) day-ahead spot prices have been studied in (Paraschiv et al., 2014) based on the Variable Renewable Energy (VRE), where the rise of the VRE penetration level tends to decrease the market spot prices but enhances the extreme spot spikes. Furthermore, the research found that although VRE decreases MCP, the effect of the FIT policy in Germany drove the prices to be higher at the end-customer side. The Turkish electricity spot market was analyzed by (Sirin and Yilmaz, 2020), where the MOE was estimated based on the impact of the wind and hydroelectric power production technologies on the spot prices. The study employed the quantile regression method, whereas the MOE of both energy technologies was significant in price reduction. However, the study recommended revising the remuneration mechanism to include the spatiotemporal values of the VRE in the electricity market reformulation. California's spot and real-time electricity markets analyzed by (Woo et al., 2016) showed that, the estimated MOE of renewable energy has a limited impact on cutting the end-user bill, while less adversely impacting the state's investment incentives for natural-gas-fired generation. The Italian day-ahead wholesale electricity market was analyzed by (Clò et al., 2015) to empirically estimate MOE from 2005 to 2013. In their empirical analysis, the MOE was calculated by multiplying the estimated coefficients of solar and wind by the weighted average load values to quantify the load reduction due to the VRE penetration. The study found that the addition of one GWh from solar and wind decreases on average the MCP price by 2.3€/MWh and 4.2€/MWh, respectively. The MOE and electricity spot market values for multi-countries were econometrically analyzed by (Welisch et al., 2016). The study assessed the VRE contribution of 73% from the European spot markets using multivariate regression analysis and ex-post calculations. Their results have shown significant static coefficients, where the VRE has shown a significant negative impact on the spot prices among all associated countries. However, the study didn't consider energy resources other than VRE in the econometric formulation. In this context, panel regression was applied to analyze the electricity prices and renewable energy in the European Union (Cech, 2016), where Hausman statistics is employed to differentiate between the fixed effect and first difference estimation methods. The German electricity spot market has been econometrically analyzed using linear and non-linear regression by (López and Nursimulu, 2019). The scope of the study is to assess the electricity price drivers. Although VRE quantities, emissions allowances, coal, gas, and oil prices are major drivers in the formulated function, the study applied the prices of the energy sources while disregarding the

Table 1
Data description.

Variable	Unit	Description	Data Source
Regional Supply	GWh	Hourly regional energy supply per production technology.	Regional electric power utilities[1]
National Demand	GWh	Hourly national energy demand	Consolidation of the data from regional electric power utilities
National Supply	GWh	Hourly national energy supply by technology	Consolidation of the data from regional electric power utilities
JEPX Trade Volume	GWh	The hourly contracted amount at the MCP	(Japan Electric Power Exchange (JEPX), 2022)
MCP	¥/kWh	JEPX system price, converted hourly by the weighted average of trade volume	(Japan Electric Power Exchange (JEPX), 2022)
Price Regional	¥/kWh	JEPX regional prices, converted hourly by the weighted average of trade volume	(Japan Electric Power Exchange (JEPX), 2022)
Average Lag Price	¥/kWh	Average lag MCP across all 24 h of the previous delivery day.	Processing of JEPX data by the authors
Lag Spot Price (***LSP***)	¥/kWh	Lag MCP for the same hour as the previous delivery day.	Processing of JEPX data by the authors
Week Spot Price (***WSP***)	¥/kWh	Lag MCP for the same hour as the previous week delivery day.	Processing of JEPX data by the authors
Spot Price Volatility	¥/kWh	Standard deviation of MCP for five previous spot prices of the typical hour.	Processing of JEPX data by the authors
LNG	1,000 ¥/Ton	Monthly cost, insurance and freight (CIF) prices of liquified natural gas.	(Ministry of Finance, Trade Statistics of Japan, 2022)
OIL	1,000 ¥/KL	Monthly cost, insurance and freight (CIF) prices of crude oil.	(Ministry of Finance, Trade Statistics of Japan, 2022)
Holiday	Dummy	Saturdays, Sundays, and all the national holidays from FY 2016–2020	Assumption: One at specified, zero at others
Daytime	Dummy	Day-ahead daytime, according to JEPX, hours between 8:00 to 22:00	Assumption: One at specified, zero at others
Year	Dummy	Fiscal year from 1st April to 31st March	Assumption: One at relevant year, zero at others

amount of injected energy for the remaining resources. On the other hand, the dynamics of hydroelectric and wind power were investigated versus the electricity prices using econometric analysis (Wen et al., 2022). Wen and his team analyzed New Zealand's Electricity market, which is a hydroelectric-dominated market. Their study found that, although the price reduction is associated with 10% of the wind penetration variation, the highest price reduction occurs in relation to the wet season as the hydroelectric penetration is at its peak. In the Japanese spot market, the first study tends to analyze the MOE of JEPX was published by (Sakaguchi and Fujii, 2021), which analyzed the MOE of the VRE on MCP, range price, and areas. The MOE is estimated from the ordinary least squares regression coefficients, which is represented as the slope difference resulting from the addition of one GWh of solar and wind. The research results revealed that the price spikes are partially controlled by solar generation, while the contribution of wind generation is not significant. Furthermore, the MOE of wind and solar power generation, in the form of an additional one GWh to the system, decreases the spot prices by 2.77 €/MWh and 0.68 €/MWh, respectively. In our study, although we aim to include the energy prices as well as all the existing energy production technologies, that would be subject to the robustness validation tests, which have strict exogeneity condition as detailed in section three of the econometric analysis.

1.4. Impact of market fragmentation on the MCP and regional spot prices

The Locational Marginal Pricing (LMP) plays a vital role in determining the market participants' bids. The LMP is a short-term approach that determines the wholesale energy prices according to the value of electric energy (i.e., losses and congestion) at various locations (Morstyn et al., 2019). Accordingly, the wholesale electricity markets are divided into zonal where the area classifications are broader than the nodal, which quantifies the locations based on close distribution nodes. Since the nodal pricing scheme is oriented to the optimal power flow models in smaller-scale communities, and the zonal pricing scheme tends to optimal planning of the energy resources among regional scales, the JEPX follows zonal pricing (Shah and Chatterjee, 2020). In JEPX, as a zonal market, the unified MCP is applicable for energy transactions nationwide. However, during the existence of the regional interconnection capacity restriction, the regional prices override the MCP for the transactions in each region (Renewable Energy Institute, 2020). The correlation of electricity prices in European wholesale power markets was empirically analyzed in (Bobinaitė et al., 2006), using the ordinary least squares regression to find the relationship between the prices, where the wholesale prices in Norway, Sweden, and Finland are lower than in other European electricity markets. The price interdependency analysis found that the rise of electricity prices in Austria and France by 1 €/MWh would raise the price in the Netherlands by 0.25 €/MWh and by 0.30 €/MWh, respectively. (Nepal, 2012) investigated the potential of market integration of the Irish Single Electricity Market (SEM) with wholesale electricity markets in Europe, using the time-varying Kalman filter technique, considering the spot prices as indices. In the study, the spot prices correlation of SEM with the other markets' spot prices indicated the integration ability with the Nord Pool Spot Market and GB European Power Exchange Market. The central Romanian electricity spot market was analyzed, using correlation analysis (Busu et al., 2019). They concluded that the higher interconnections between the region's countries, the lower price volatilities strikes are present. In the JEPX, besides their MOE analysis, Sakaguchi and Fujii have analyzed the regional price correlations using Ordinary Least Square (OLS) regression, whereas the proposed methodology employs the goodness-of-fit measure (R^2) as a correlation significance index for the regional price. This methodology assigns a regional price as a singular dependent variable against another regional price as a singular independent variable. The limitation of their methodology is that it restricts the impact of each regional price on the cross-bonding regional price while ignoring any other potential independent variable. Although our study employs the OLS regression to analyze the regional prices, the two-stage least square method is developed to analyze the regional prices interdependency by considering various independent variables, including energy resources price and injected electric power quantities.

1.5. Motivations and contributions

Based on the literature survey, although the merit order effect considering the impact of variable renewable energy technologies is widely investigated in earlier research contributions, it lacks a comprehensive econometric analysis of existing energy production technologies. From the supply curve perspective, the major energy pricing influencers are the contribution of each energy production technology and the physical locations among the market structure. For zonal pricing markets such as JEPX, each region has dedicated supply resources and special demand patterns. Sequentially, when the regional supply does not meet the regional demand, the regional prices override the MCP. Hence, the impact of energy production technologies

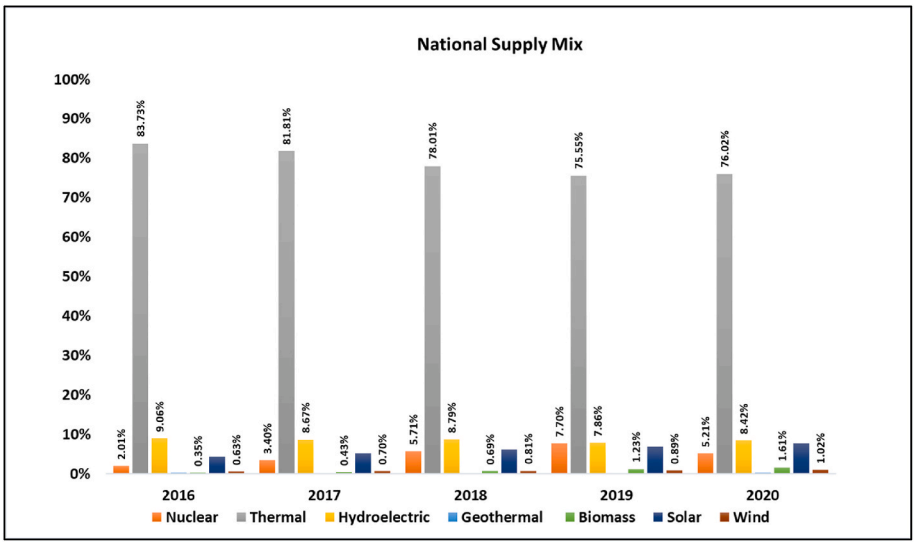


Fig. 2. National supply mix per technology based on the installed capacity from 2016 to 2020.

diversification is appreciable at the JEPX in the form of a national competitive market with singular MCP, through a fragmented market with regional prices. On the other hand, although the current active liberalization process in JEPX makes it one of the rapidly evolving electric power exchange markets globally, it is still lacking in the literature and needs to be further investigated. The study offers threefold core contributions as follows:

- A detailed quantitative analysis of the energy policy regulations in the JEPX market.
- Quantitative estimation of the merit order effect on MCP based on various energy production technologies and the fragmented spot market prices.
- Determining the interdependency of the fragmented spot prices considering MCP, market trading volume, and regional interconnections.

2. Data sources

The historical data acquisition was conducted from several sources as listed in Table 1. The data collection has begun with the cross-sectional spot electricity prices of the nine TSOs[1] associated with JEPX, while the system price is the tenth cross-section as the system MCP. The regional and system prices are collected from (Japan Electric Power Exchange (JEPX), 2022) in 30 min intervals, from 2016 to 2020. To align the time dimension of the panel data, the 30 min intervals have been converted to hourly records using weighted averages based on the market trading volume. The regional supply production and the regional demand have been collected from the regional TSO, whereas the national supply and demand patterns are obtained through regional data consolidation. Each regional TSO provides the hourly amount of the region's demand, thermal power generation (Th), nuclear power generation (Nu), hydroelectric power generation (Hy), geothermal power generation (Geo), biomass power generation (Bio), photovoltaic solar power generation (PV), wind power generation (WP), Hydroelectric storage power cycles (HS), and regional interconnection power (Con). The weekends and the national public holidays are employed as the first dummy variable, while the daytime of the JEPX pricing scheme and fiscal years are the second and third dummy variables, respectively.

The acquired data were arranged into two formats. First, the long panel dataset format, which has a vertical dataset arrangement. In this arrangement, each year has 8,760 h except the leap year 2020 with 8,784 h. Thus, each regional cross-section has 43,824 hourly records, which forms a vertical representation of 438,240 data records of the nine regions' spot prices according to quantities supplied. The national cross-section consolidates the regional quantities with respect to the MCP. Second, wide time-series dataset format, with a horizontal data arrangement, forms 43,824 hourly records. In this format, the MCP, regional spot prices, regional supply, and demand quantities, and JEPX trading volume are represented horizontally, whereas each explanatory variable is assigned as an independent index.

2.1. Exploratory variables driving the model

As motivated in section 2, the study analyzes the MCP from the supply resources perspective. The MCP and market trading volume are determined based on supply and demand curves, where supply technologies identify the quantity capabilities and the cost (Farzaneh, 2019). The first part of the study scope is a quantitative analysis of the MCP and the regional spot prices based on the regional supply curves and the consolidated national supply curve. The second part is a quantitative estimation of the regional spot prices and MCP correlations, and consequently, the regional spot price interdependency relation when other spot prices are subjected to change. Thus, the study considers the fragmented supply stack as the core exploratory variables driving the model prices.

2.1.1. Supply stack variables

Japan's national electricity supply mix is dynamic under the Long-term Energy Supply and Demand Outlook for the years prior to 2030 (METI, 2015). Nuclear power is planned to supply the baseload with a share of 20%–22%, whereas the nuclear share before the Great East Japan Earthquake was 25%. The thermal generation is expected to dominate 56% of the national supply with 27% Liquefied Natural Gas (LNG), 26% coal, and 3% oil. Renewable energy is projected to share 22%–24%, with hydro 9%, geothermal 1.1%, biomass 3.7%–4.6%, wind 1.7%, and solar 7% (METI, 2015). Based on the collected data (JEPX, 2022), Fig. 2 demonstrates the dynamicity of the supply diversification

[1] 1. (Chubu electric power grid, 2022); 2. (Chugoku electric power transmission and distribution company, 2022); 3. (Hokkaido electric power network, 2022); 4. (Hokuriku electric power transmission and distribution company, 2022); 5. (Kansai transmission and distribution, 2022); 6. (Kyushu electric power transmission and distribution, 2022); 7. (Shikoku electric power transmission and distribution company, 2022); 8. (TEPCO power grid, 2022); 9. (Tohoku electric power network, 2022).

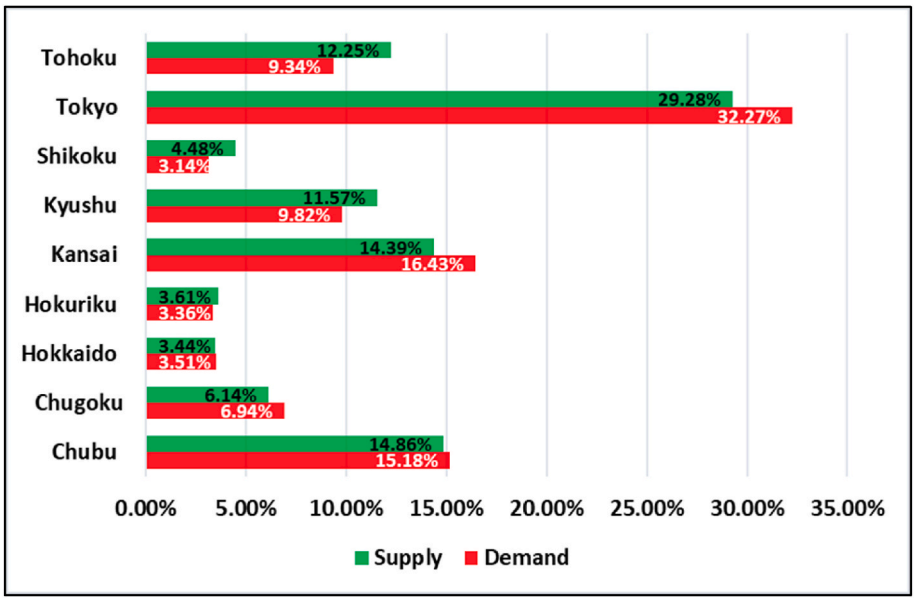


Fig. 3. Regional supply and demand balance on an average basis (2016–2020).

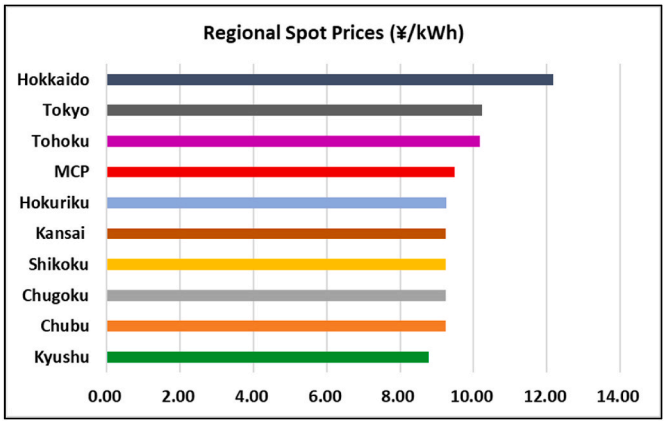


Fig. 4. Regional hourly day-ahead market prices average basis (2016–2020).

between 2016 and 2020.

Considering the regionally fragmented supply and demand balance, the energy cycles of hydroelectric storage and the regional interconnections are vital in determining the regional spot prices and the MCP. Although the MCP is applicable in nationwide energy transitions, the regional spot prices are imposed with the market fragmentation. Fig. 3 depicts the average share of the regional supply and demand patterns from the national base. The regional spot prices vary among regions due to the regional supply and demand balance, which is reflected in the interconnected regions. Fig. 4 shows the average regional variation of spot prices from 2016 to 2020, whereas the MCP is found between the regions.

2.1.2. Historical price and risk variables

The comprising of the lagged MCP for the same hour of the previous delivery day, the same hour of the typical day a week before, and the average MCP for the 24 h before the relevant day helps to reduce the serial correlation effect in the data. In addition, the spot price volatility is included to measure the price instability, which is computed using the standard deviation of the spot prices for the same hour of the last five relevant days (Paraschiv et al., 2014). The lagged prices and price volatility index are employed in JEPX by (Sakaguchi and Hidemichi, 2021). The necessity to use the historical price and risk variables stems from the adaptive behavior of the spot prices sensitivity to the lagged prices. The adaptive behavior reinforces the previous successful bids, which duplicates the sequence of the high prices. Thus, a positive sensitivity signal to the lagged price is expected when repeated gaming elements in the market are present (Bunn et al., 2016).

3. Econometric analysis

The study employs two linear regression models to examine the impact of market liberalization on the spot electricity price in the Japanese electric power exchange market. The stream of the section context is summarized in the overall methodology flowchart of Fig. 5, where the first model of the merit order effect econometric uses the quantities of energy supply stack variables to analyze the panel data and estimate the pricing impact (Farzaneh, 2019), using Fixed Effect Two-Stage Least Squares (FE2SLS) estimator (Wooldridge, 2010). The historical data are arranged in an extended panel dataset format, where the MCP and regional spot prices are assigned as dependent variables. On the other hand, the second model is the regional prices interdependency model, which is divided into two sub-models. The first sub-model is correlation analysis to the JEPX prices using Ordinary Least Square (OLS) as applied in (Sakaguchi and Hidemichi, 2021), where the goodness-of-fit measure (R^2) is used to explain the significance of the relationships among two spot prices. The second sub-model employs a two-stage least square (2SLS) estimator to find the regional spot price interdependency on MCP. The spot price interdependency concept has been adopted from (Bobinaitė et al., 2006). The same historical data are in a wide dataset format, which assigns each regional explanatory variable to an independent data index (Wooldridge, 2010).

3.1. The merit order effect econometric model

The goal of the proposed model is to determine the impact of changing in one unit production per supply technology on the spot price. The spot price building mechanism by JEPX is determined through synthesizing the merit order curve of the supply quantity-price, which vitally relies on each supply technology. The study fits a panel linear regression model to estimate the spot price variation according to the change of supply production per technology and consider the historic price with risk signals as expressed in equation (1):

$$y_{i,t} = \beta_o + \beta_{k1} x_{i,t} + \beta_{k2} w_{i,t} + c_i + u_{i,t} \quad (1)$$

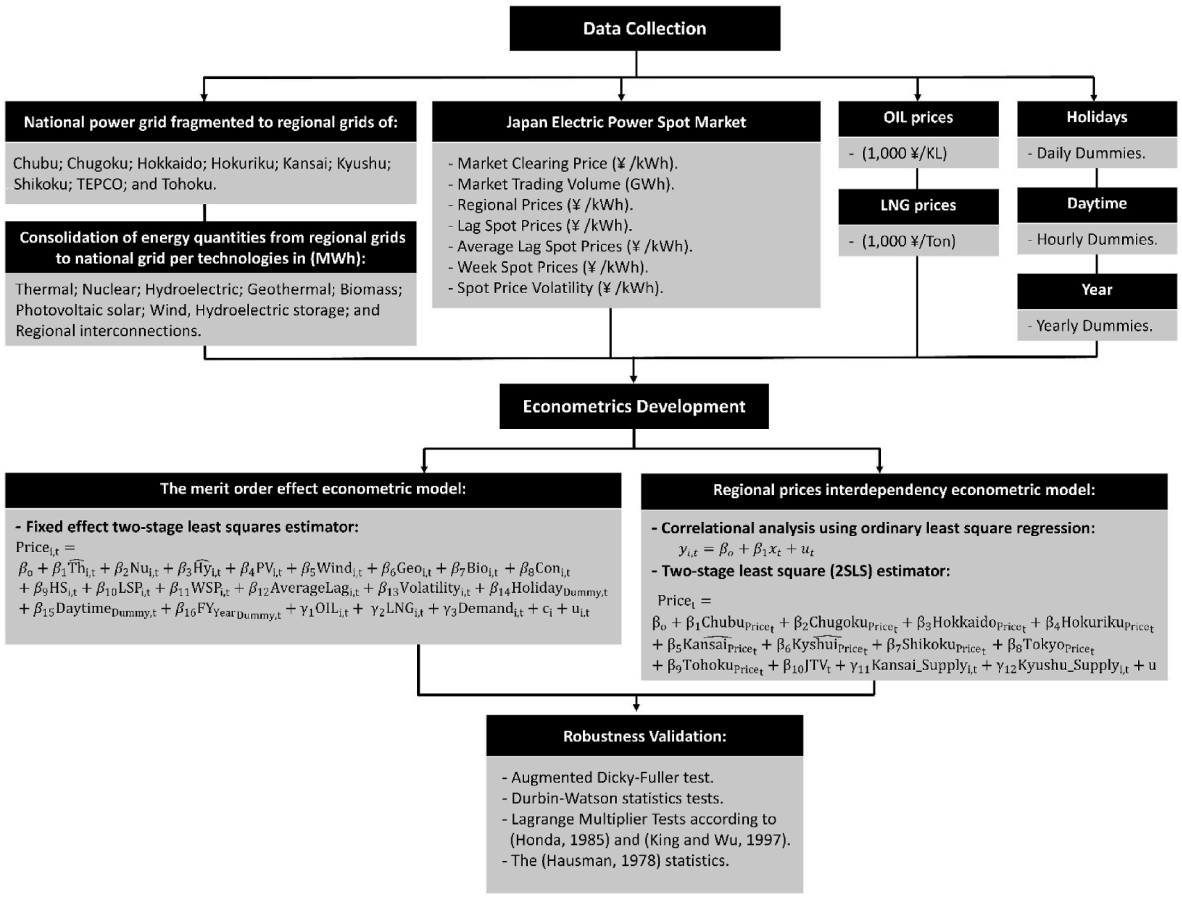


Fig. 5. Overall flowchart of the proposed methodology.

Where $y_{i,t}$ is the $T \times 1$ vector of dependent variables that denotes the spot price for each region i per hour t. The independent variable $x_{i,t}$ is a $T \times K_1$ matrix, which denotes the quantity supplied x per the energy technologies K_1 (i.e., Th, Nu, Hy, Geo, Bio, PV, WP, HS, Con) at each region i per hour t for the total observations N of 438,240. The second independent variable $w_{i,t}$ is a $T \times K_2$ matrix denotes the lag values as in Table 1 in monetary units per measurement methodology K_2 (i.e., *LSP*, *LWP*, Average Lag_t, Volatility_t) at each i per t for the total number of observations N. The unobserved effect c_i captures the individual fixed effect of the cross-sections i, while $u_{i,t}$ is the idiosyncratic error that captures the changes across t and i. The FE estimator consistency is validated on the orthogonality condition assumption (1.2), whereas all the explanatory variables of $x_{i,t}$ and $w_{i,t}$ are under strict exogeneity conditions on c_i. Although all the explanatory variables satisfied assumption (2), $Th_{i,t}$ and $Hy_{i,t}$ are endogenous variables.

$$E(u_{i,t}|x_i, c_i) = 0, t = 1, 2, \ldots T. \tag{2}$$

The Instrumental Variables (IV) are employed to estimate the parameters. The selected instrumental variables are the $OIL_{i,t}$ and $LNG_{i,t}$. In addition, the $Demand_{i,t}$ is employed as IV since the endogenous variables are oriented to the baseload. The IVs are denoted $z_{i,t}$ which is $T \times K_3$ matrix, while K_3 (i.e., *OIL*, *LNG*, *Demand*) at each i per t for the total observations N. The consistency assumption of the panel regression is applied for the selected IVs, that including (3) where strict exogeneity of the instruments on c_i is assured.

$$E(u_{i,t}|z_i, c_i) = 0, t = 1, 2, \ldots N. \tag{3}$$

The reduced form equation for endogenous variables is expressed by substituting $x_{i,t}$ exogenous variables with respect to K_1, $w_{i,t}$ exogenous variables with respect to K_2, and the instruments $z_{i,t}$ with respect to K_3 from the structural equation (1) in (4) and (5). The fitted values of the first stage regression $\widehat{Th}_{i,t}$ and $\widehat{Hy}_{i,t}$ is obtained according to the two-stage least square method.

$$\widehat{\text{Th}}_{i,t} = \delta_o + \delta_{01}\text{Nu}_{i,t} + \delta_{02}\text{HY}_{i,t} + \delta_{03}\text{PV}_{i,t} + \delta_{04}\text{Wind}_{i,t} + \delta_{05}\text{Geo}_{i,t} + \delta_{06}\text{Bio}_{i,t} + \delta_{07}\text{Con}_{i,t} + \delta_{08}\text{HS}_{i,t} + \delta_{09}\text{LSP}_{i,t} + \delta_{010}\text{WSP}_{i,t} + \delta_{011}\text{AverageLag}_{i,t} + \delta_{012}\text{Volatility}_{i,t} + \delta_{013}\text{Holiday}_{\text{Dummy},t} + \delta_{014}\text{Daytime}_{\text{Dummy},t} + \delta_{015}\text{FY}_{\text{Year Dummy},t} + \theta_{01}\text{OIL}_{i,t} + \theta_{02}\text{LNG}_{i,t} + \theta_{03}\text{Demand}_{i,t} + c_{i0} + u_{i,t0} \tag{4}$$

$$\widehat{\text{Hy}}_{i,t} = \delta_{o1} + \delta_{11}\text{Nu}_{i,t} + \delta_{12}\text{Th}_{i,t} + \delta_{13}\text{PV}_{i,t} + \delta_{14}\text{Wind}_{i,t} + \delta_{15}\text{Geo}_{i,t} + \delta_{16}\text{Bio}_{i,t} + \delta_{17}\text{Con}_{i,t} + \delta_{18}\text{HS}_{i,t} + \delta_{110}\text{LSP}_{i,t} + \delta_{111}\text{WSP}_{i,t} + \delta_{112}\text{AverageLag}_{i,t} + \delta_{113}\text{Volatility}_{i,t} + \delta_{114}\text{Holiday}_{\text{Dummy},t} + \delta_{115}\text{Daytime}_{\text{Dummy},t} + \delta_{115}\text{FY}_{\text{Year Dummy},t} + \theta_{11}\text{OIL}_{i,t} + \theta_{12}\text{LNG}_{i,t} + \theta_{13}\text{Demand}_{i,t} + c_{i1} + u_{i,t1} \tag{5}$$

By substituting the reduced form equations (4) and (5) into the structural equation (1), and replacing the IV fitted coefficients θ_k by the new estimates γ_k, the rearranged reduced form is obtained in equation (6).

$$\begin{aligned}\mathrm{Price}_{i,t} = {} & \beta_o + \beta_1 \widehat{\mathrm{Th}}_{i,t} + \beta_2 \mathrm{Nu}_{i,t} + \beta_3 \widehat{\mathrm{Hy}}_{i,t} + \beta_4 \mathrm{PV}_{i,t} + \beta_5 \mathrm{Wind}_{i,t} + \beta_6 \mathrm{Geo}_{i,t} + \beta_7 \mathrm{Bio}_{i,t} + \beta_8 \mathrm{Con}_{i,t} + \beta_9 \mathrm{HS}_{i,t} + \\ & \beta_{10} \mathrm{LSP}_{i,t} + \beta_{11} \mathrm{WSP}_{i,t} + \beta_{12} \mathrm{AverageLag}_{i,t} + \beta_{13} \mathrm{Volatility}_{i,t} + \beta_{14} \mathrm{Holiday}_{\mathrm{Dummy},t} + \beta_{15} \mathrm{Daytime}_{\mathrm{Dummy},t} + \\ & \beta_{16} \mathrm{FY}_{\mathrm{YearDummy},t} + \gamma_1 \mathrm{OIL}_{i,t} + \gamma_2 \mathrm{LNG}_{i,t} + \gamma_3 \mathrm{Demand}_{i,t} + c_i + u_{i,t}\end{aligned} \tag{6}$$

Given the variables of (6) in a linear format, the estimated coefficients of each independent variable indicate the slope, where the dependent variable increases as a response to one unit increase in a particular dependent variable. The obtained slope is specifically associated with the changing variable, while the remaining independent variables are constants, as indicated in equation (7). Therefore, the MOE is the shift in the spot price resulting from an additional gigawatt-hour of electricity per technology in a given hour. The MOE for each energy technology is represented by the estimated coefficients α_{k1} , as follows:

$$\beta_{k1_*} = \frac{\Delta y_{i,t_*}}{\Delta x_{i,t_*}} \tag{7}$$

3.2. The regional prices econometric model

This model performs a correlational analysis that determines the correlation between the MCP and the regional spot prices, considering the JEPX traded volume. The first part of this model uses simple OLS where R^2 indicates the efficiency of the relation between two spot prices (i.e., MCP and regional spot price, or two regional spot prices and vice versa) formed from equation (8). The model consistency is validated under strict exogeneity condition of assumption (9)

$$y_{i,t} = \beta_o + \beta_1 x_t + u_t \tag{8}$$

$$E(u_t) = 0, \quad \mathrm{Cov}(x_t, \mathrm{u}) = 0, t = 1, 2, \ldots T. \tag{9}$$

Where y_t is the $T \times 1$ vector of dependent variables that denotes the spot price per time t. The independent variable x_t is a $T \times K$ matrix which denotes the adverse spot price of the other region or the MCP per hour t for the total observations (N) of $43,824$.

In the second part, the model estimates the MCP interdependency relation with each region. Since the MCP is always applicable in the energy transactions nationwide unless during market fragmentation presence. In JEPX, market fragmentation occurs when regional interconnection limitation exists (e.g., when demand and supply at different regions with interconnection restriction, the prices are calculated for each region). Therefore, this model tends to analyze the impact of change at the regional spot price on the MCP as expressed in equation (10):

$$\begin{aligned}\mathrm{Price}_t = {} & \beta_o + \beta_1 \mathrm{Chubu}_{\mathrm{Price}\,t} + \beta_2 \mathrm{Chugoku}_{\mathrm{Price}\,t} + \beta_3 \mathrm{Hokkaido}_{\mathrm{Price}\,t} + \beta_4 \mathrm{Hokuriku}_{\mathrm{Price}\,t} + \beta_5 \mathrm{Kansai}_{\mathrm{Price}\,t} + \\ & \beta_6 \mathrm{Kyushu}_{\mathrm{Price}\,t} + \beta_7 \mathrm{Shikoku}_{\mathrm{Price}\,t} + \beta_8 \mathrm{Tokyo}_{\mathrm{Price}\,t} + \beta_9 \mathrm{Tohoku}_{\mathrm{Price}\,t} + \beta_{10} \mathrm{JTV}_t + u\end{aligned} \tag{10}$$

Where Price_t is the $T \times 1$ vector of dependent variables that denotes the MCP per time t. The independent variables denote the spot price for each region per t for the total observations of $N = 43824$. The JTV_t is the JEPX trading volume per t for the total number of observations T. The dummy variable holiday is indicated by one for weekends and public holidays and zero for others, while the daytime dummy as specified by JEPX is indicated by one for hours between 8 and 22 and zero for others. The year dummies are represented fiscally starting from 1st April to the 31st of March. The model consistency is validated under the strict exogeneity condition of the orthogonality condition assumption (1.11), in which the error term u is correlated with none of the explanatory variables. However, the null hypothesis is not rejected for Kansai and Kyushu prices which were found to be endogenous variables:

$$\mathrm{E(u)} = 0, \qquad \mathrm{Cov(independent\ \ variables_t, u)} = 0, \qquad \mathrm{t} = 1, 2, \ldots 43824 \tag{11}$$

The regional supply mix, which includes the available production technologies (i.e., Th, Nu, Hy, Geo, Bio, PV, WP, HS, Con) for the regions of endogenous variables, are validated on assumption (1.12) to be employed as instrumental variables.

$$E\left(u_{i,t} \middle| z_i, c_i\right) = 0, t = 1, 2, \ldots N. \tag{12}$$

The reduced form equation of Kansai and Kyushu spot prices are obtained in (1.13) and (1.14), respectively.

$$\begin{aligned}\widehat{\mathrm{Kansai}}_{\mathrm{Price}\ t} = {} & \delta_o + \delta_1 \mathrm{Chubu}_{\mathrm{Price}\ t} + \delta_2 \mathrm{Chugoku}_{\mathrm{Price}\ t} + \delta_3 \mathrm{Hokkaido}_{\mathrm{Price}\ t} \\ & + \delta_4 \mathrm{Hokuriku}_{\mathrm{Price}\ t} + \delta_6 \mathrm{Kyushu}_{\mathrm{Price}\ t} + \delta_7 \mathrm{Shikoku}_{\mathrm{Price}\ t} + \delta_8 \mathrm{Tokyo}_{\mathrm{Price}\ t} \\ & + \delta_9 \mathrm{Tohoku}_{\mathrm{Price}\ t} + \delta_{10} \mathrm{JTV}_t + + \theta_{11} \mathrm{Kansai_Supply}_{i,t} \\ & + \theta_{12} \mathrm{Kyushu_Supply}_{i,t} + u\end{aligned} \tag{13}$$

$$\begin{aligned}\widehat{\mathrm{Kyushu}}_{\mathrm{Price}\ t} = {} & \delta_o + \delta_1 \mathrm{Chubu}_{\mathrm{Price}\ t} + \delta_2 \mathrm{Chugoku}_{\mathrm{Price}\ t} + \delta_3 \mathrm{Hokkaido}_{\mathrm{Price}\ t} \\ & + \delta_4 \mathrm{Hokuriku}_{\mathrm{Price}\ t} + \delta_6 \mathrm{Kansai}_{\mathrm{Price}\ t} + \delta_7 \mathrm{Shikoku}_{\mathrm{Price}\ t} + \delta_8 \mathrm{Tokyo}_{\mathrm{Price}\ t} \\ & + \delta_9 \mathrm{Tohoku}_{\mathrm{Price}\ t} + \delta_{10} \mathrm{JTV}_t + + \theta_{11} \mathrm{Kansai_Supply}_{i,t} \\ & + \theta_{12} \mathrm{Kyushu_Supply}_{i,t} + u\end{aligned} \tag{14}$$

The rearranged reduced form equation is obtained by plugging (13) and (14) into (10).

$$\begin{aligned}\mathrm{Price}_t = {} & \beta_o + \beta_1 \mathrm{Chubu}_{\mathrm{Price}\ t} + \beta_2 \mathrm{Chugoku}_{\mathrm{Price}\ t} + \beta_3 \mathrm{Hokkaido}_{\mathrm{Price}\ t} \\ & + \beta_4 \mathrm{Hokuriku}_{\mathrm{Price}\ t} + \beta_5 \widehat{\mathrm{Kansai}}_{\mathrm{Price}\ t} + \beta_6 \widehat{\mathrm{Kyshui}}_{\mathrm{Price}\ t} + \beta_7 \mathrm{Shikoku}_{\mathrm{Price}\ t} \\ & + \beta_8 \mathrm{Tokyo}_{\mathrm{Price}\ t} + \beta_9 \mathrm{Tohoku}_{\mathrm{Price}\ t} + \beta_{10} \mathrm{JTV}_t + \gamma_{11} \mathrm{Kansai_Supply}_{i,t} \\ & + \gamma_{12} \mathrm{Kyushu_Supply}_{i,t} + u\end{aligned} \tag{15}$$

3.3. Robustness validation

To determine the stationarity of the time series variables of JEPX spot prices, the robustness of the data is checked using Augmented Dicky-Fuller (ADF) test (Dickey and Fuller, 1979). The applied test has rejected the null hypothesis of a unit root at a 1% significance level. The

serial correlation in idiosyncratic errors has been tested using Durbin-Watson statistics, and the null hypothesis is rejected at a significance of 1%, whereas the use of lagged data is feasible to reduce serial correlation. Lagrange Multiplier Tests is performed for FE estimator according to (Honda, 1985) and (King and Wu, 1997), and the null hypothesis is rejected at the significance of 1%. The (Hausman, 1978) statistics are performed to validate the instruments, where the null hypothesis for all the scope variables is being exogenous with a significantly less than 1%.

4. Results and discussion

Table 2 presents the slope estimates of each energy production technology, where a one GWh addition from nuclear, hydroelectric, geothermal, biomass, solar, or wind power tend to decrease the spot prices. Contrarily, adding one GWh from thermal, hydroelectric storage, or interconnections raises the spot prices. Although the estimated price reduction resulting from renewable energy production technologies is expected, the slight MCP reduction from nuclear energy production is likely to persist due to the gradual resumption since Great East Japan Earthquake.

The highest MCP influencer is geothermal power. However, the low contribution to the national energy mix below 1%, and the future directions to increase the penetration to 1.1% by 2030 make the geothermal impact slender. The MOE of one GWh addition from hydroelectric power on the MCP is 769 ¥/MWh, where the penetration level at the national demand is planned to persist until 2030, as discussed in section 3.1.1. The addition of one GWh from biomass electric power is estimated to lower the MCP by 295 ¥/MWh; thus, the METI intends to raise the share of this technology to around twofold by 2030. The hydroelectric storage indicates the highest marginal cost among the technologies with 769 ¥/MWh for each additional GWh. On the other hand, the spot prices are higher during the daytime and lower during weekends and national public holidays.

The estimated price effect of 2017 and 2018 are approximately equal, that is due to the indiscernible major deregulation activities in the market during the time. In 2019, a significant spot price drop has started due to the expiration of the 10 years fixed rates contracts of the surplus solar power purchasing system, which met with FIT policy in effect, that reflected on yearly estimates of Table 2 as the least price increment among the subject years compared to the base of 2016. Given this, the MCP was dropped to near 1 ¥/kWh and spiked up to 60 ¥/kWh for a few spots in response to the surplus solar power purchasing provider's involvement in the FIT scheme. Consequently, the MCP and price volatilities were dropped to the least, as depicted in Fig. 6. The slight MCP increments from Fig. 6 of 2017 and 2018 are positively translated in the yearly estimates of Table 2.

Fig. 7 demonstrates the regional spot prices against the regional demand plotted according to VRE penetration levels. The penetration level is calculated via the ratio of summing the hourly solar and wind produced power by the hourly demand. The spot prices tend to spike when the penetration level is at the minimum level, whereas no price spikes are detected when the penetration level is higher than 50%. Given this, and considering VRE intermittency, the hydroelectric storage cycles of 2016–2019 in Fig. 8 shows typical crescent moon-shaped patterns, where the highest charging cycle is found below 25 ¥/MWh. However, in addition to the stochastic distribution of the hydroelectric storage cycles in 2020, the charging cycles escalate throughout the spots up to 200 ¥/MWh. The penetration level of hydroelectric storage cycles is calculated via the ratio of hourly national hydroelectric storage share by the hourly national supply. Fig. 9 shows that the spot price spikes higher than 100 ¥/kWh have been detected from 2020 onwards. This is justified due to the increase of the annual average VRE penetration with respect to the yearly average hydroelectric cycles penetration shown in Fig. 10. Sequentially, corresponding to the highest VRE penetration in 2020 depicted in Fig. 11, the MCPs significantly dropped to near zero and spiked up to 250 ¥/kWh for considerable spots in all regions. Consequently, each region is restricted to balance the instantaneous intermittency of the VRE through the local hydroelectric storage as the fastest responding technology, which leads to higher duties of local hydroelectric storage instead of using the interregional connections.

The MOE of the VRE obtained in the study has been compared with similar contributions, as shown in Table 3. The obtained MOE varies compared to (Sakaguchi and Hidemichi, 2021) due to the significant price volatilities of 2020. In addition, the VRE penetration level overran 50% of the national supply at frequent spots, as demonstrated in Fig. 11, which led to higher duties of hydroelectric storage as demonstrated in Fig. 10. In the (Sakaguchi and Hidemichi, 2021) study, the covered

Table 2
Results of panel data regression.

Variable	Estimate	Std. Error	z-value	Pr(>|z|)
PV	−0.02782	0.002696	−10.3167	<2.2e-16 ***
WT	−0.75468	0.061884	−12.1949	<2.2e-16 ***
NU	−0.04369	0.011806	−3.7005	0.0002152 ***
TH	0.027402	0.001918	14.2893	<2.2e-16 ***
BIO	−0.29539	0.084001	−3.5165	0.0004372 ***
HY	−0.76941	0.088787	−8.6658	<2.2e-16 ***
HS	0.606043	0.026957	22.482	<2.2e-16 ***
CON	0.103268	0.012309	8.3897	<2.2e-16 ***
GEO	−5.83994	1.119469	−5.2167	1.821e-07 ***
LSP	0.674917	0.001532	440.4653	<2.2e-16 ***
SPV	0.139359	0.003178	43.857	<2.2e-16 ***
WSP	0.041789	0.001131	36.9584	<2.2e-16 ***
AVGSP	0.180822	0.001806	100.1154	<2.2e-16 ***
Holiday	−1.39044	0.01844	−75.4038	<2.2e-16 ***
Daytime	0.561449	0.022289	25.19	<2.2e-16 ***
Year 2017	0.211797	0.027408	7.7276	1.096e-14 ***
Year 2018	0.229083	0.028365	8.0761	6.684e-16 ***
Year 2019	0.085932	0.0296	2.9031	0.0036950 **
Year 2020	0.171554	0.034194	5.0171	5.246e-07 ***
Total Sum of Squares	55214000	Residual Sum of Squares	11554000	
R-Squared	0.79075	Adj. R-Squared	0.79074	
p-value	<2.22E-16			

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

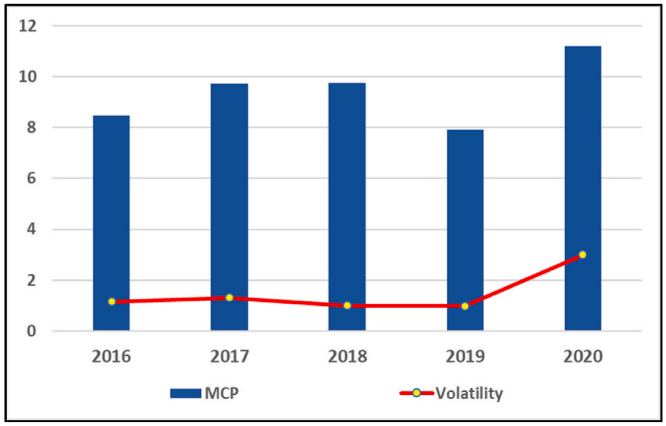


Fig. 6. The yearly MCP and price volatility (¥/kWh) average basis (2016–2020).

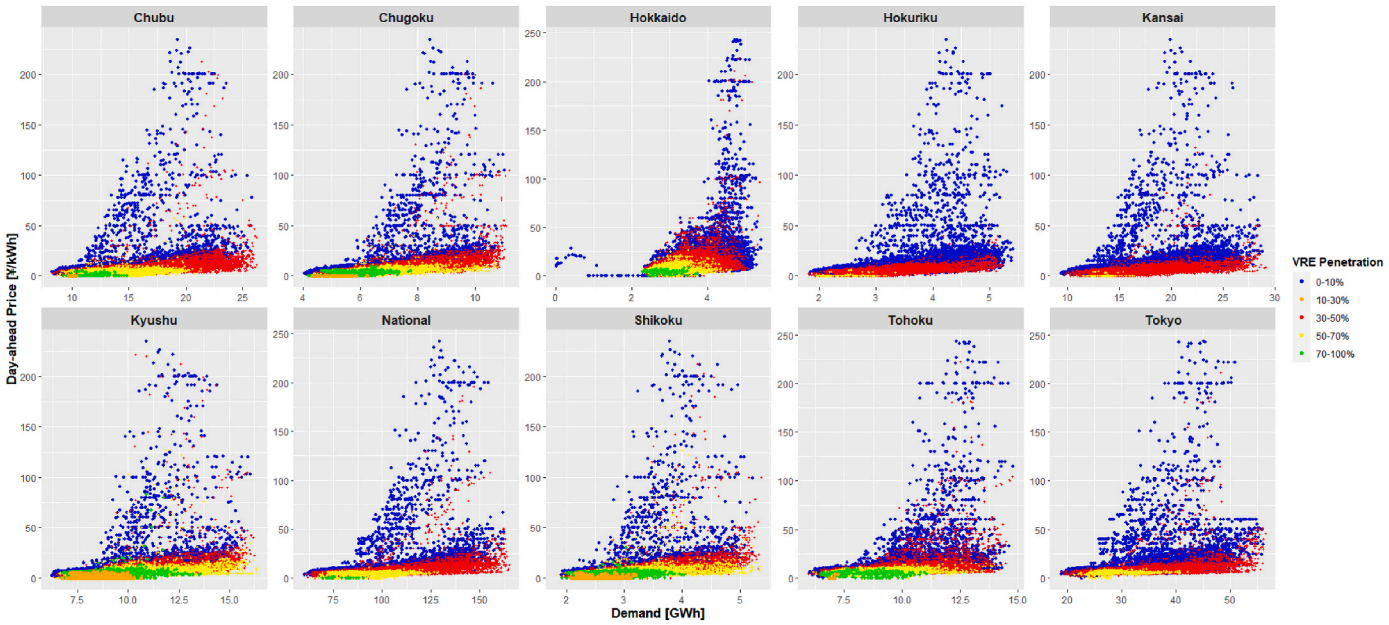


Fig. 7. Spot prices against hourly demand plotted according to VRE penetration.

period is 2016–2019.

The regional correlational analysis presented in Table 4 shows the significance of the interregional spot prices and the MCP. Hokkaido price shows the least interrelation with the MCP and the regional spot prices. The spot prices of Kansai, Shikoku, Chugoku, and Hokuriku show a highly significant correlation among the spot prices, which is justified by the presence of direct interconnections between Kansai and the remaining. Kyushu price has a high significant value correlated with most regions except Hokkaido, Tokyo, and Tohoku. Tokyo and Tohoku prices show high correlations among each other, while slightly different from the remaining regions. Consequently, the MCP shows a high correlation balance between all the regions. However, due to the geographical location of Hokkaido, the MCP has the most negligible correlation.

Table 5 shows the MCP interdependency relation with each region's prices. The estimates show the impact of adding one ¥/kWh to the region spot price on the MCP. The addition of Hokuriku and Chugoku spot prices significantly decreases the MCP. However, according to Table 5 and considering the regional interconnection between Hokuriku, Chugoku, and Kansai, the high rate influencing the MCP from Kansai is balanced by Hokuriku and Chugoku, which is correspondingly validated in Table 4. Furthermore, the high VRE penetration level at Chugoku and the low demand of both regions, as depicted in Fig. 6, are linked to justify the balance of the three region's estimates in Table 5.

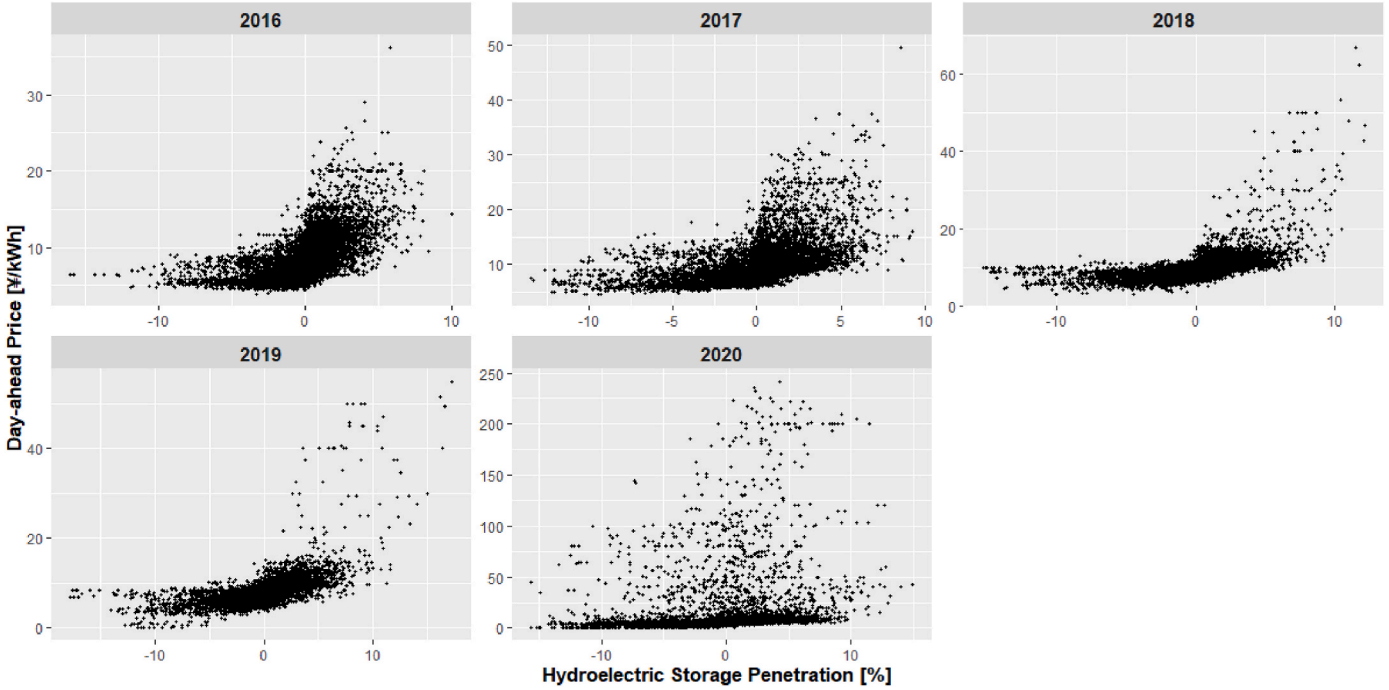


Fig. 8. MCP against hourly national hydroelectric storage penetration.

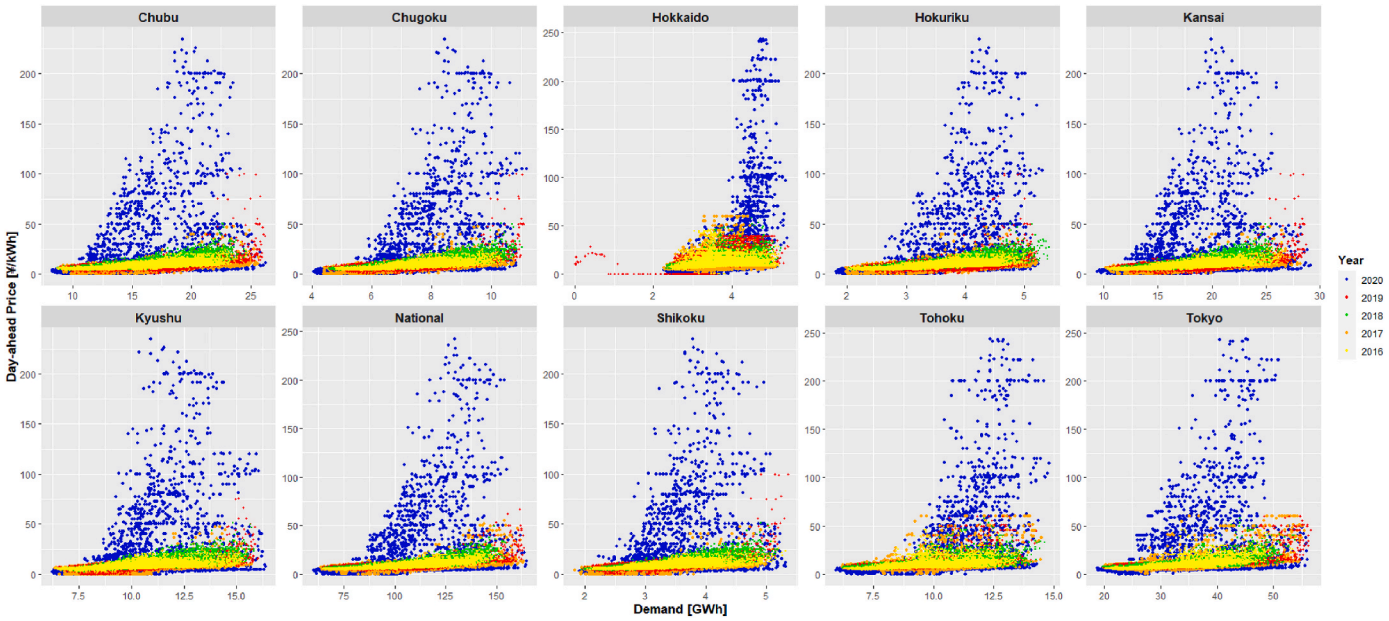


Fig. 9. Spot prices against hourly demand plotted in different years.

On the other hand, Kyushu and Tohoku have the highest potential to influence the MCP, which is justified due to supply and demand balance as demonstrated in Fig. 4 and the high VRE penetration level depicted in Fig. 6. The addition of one ¥/kWh to Tohoku price is estimated to raise the MCP by 0.35 ¥/kWh, while the addition of one ¥/kWh to Tokyo price is estimated to increase the MCP by 0.04 ¥/kWh. Thus, the price at Tohoku has a higher potential to influence the MCP than Tokyo price, which describes the correlations of Table 4 and Fig. 4, whereby Tohoku supplies the shortage demand in Tokyo through inter-regional connection. The diagnostic tests values of this model are depicted in Table A1 in the appendix.

5. Conclusion and policy implications

The study provided a historical review of Japan's electric power market liberalization. In addition, the study collected the historical data of Japan's national grid from the regional TSO, and JEPX spot prices.

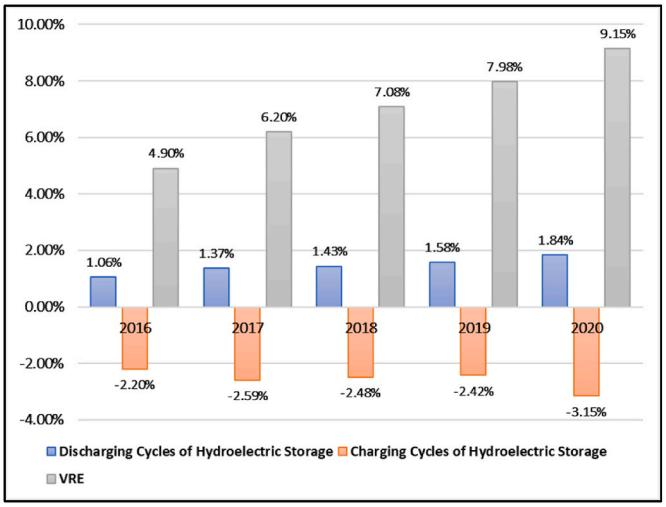


Fig. 10. The yearly penetration of hydroelectric storage and VRE (%).

The data has been analyzed using two econometrics to justify the MOE and the regional prices interdependency from the MCP. The effectiveness of the results has been validated against the Long-term Energy Supply and Demand Outlook Policy for the years prior to 2030, where the impact of adding one GWh from hydroelectric power is found to decrease the MCP by 769 ¥/MWh, while the addition of one GWh from biomass electric power is estimated to lower the MCP by 295 ¥/MWh. On the other hand, one GWh of nuclear energy and biomass energy decrease the MCP by 43.7 and 295 ¥/MWh, respectively. Contrarily, the additional GWh from thermal power hydroelectric storage or regional inter-connections raises the MCP by 27.4, 606, 103 ¥/MWh, respectively. The slight MCP reduction from nuclear energy production is expected to persist due to the gradual resumption since the Great East Japan Earthquake and until reaching the planned capacity of 20%–22%.

The price drop occurred to the MCP in 2019, and the MCP volatility dropped to near 1 ¥/kWh and spiked up to 60 ¥/kWh for a few spots is justified as an effect of the surplus solar power purchasing provider's involvement in the FIT scheme. The significant price increase and volatility dropped to near zero and spiked up to 250 ¥/kWh for considerable spots in 2020 at all regions are justified due to the increased penetration of hydroelectric storage, whereby each region is enforced to balance the instantaneous intermittency of the VRE through the local hydroelectric storage. The regional correlation analysis has shown that the spot prices at Kansai, Shikoku, Chugoku, and Hokuriku are significantly directly related, where the effect of Kansai on the MCP is balanced through the direct interconnections of Hokuriku and Chugoku. In addition, Tokyo and Tohoku spot prices show a high direct correlation, while slightly different from the remaining regions. Furthermore, Kyushu and Tohoku have reasonable control over the MCP, and that is due to the high VRE penetration and the positive balance between supply and demand patterns.

The crucial policy implications observed in this study are the various price interdependency between the regions which indicates the variation between the supply and demand patterns regionally, and the implicit auction approach, which restricts the use of the regional connections to be finalized in the spot market as day-ahead contracts. This can be a major cause of the vanished crescent moon-shaped patterns of Fig. 8 in 2020, where the spot price spiked up to 250 ¥/kWh. The crescent moon-shaped patterns from 2016 to 2019 are indicators of the

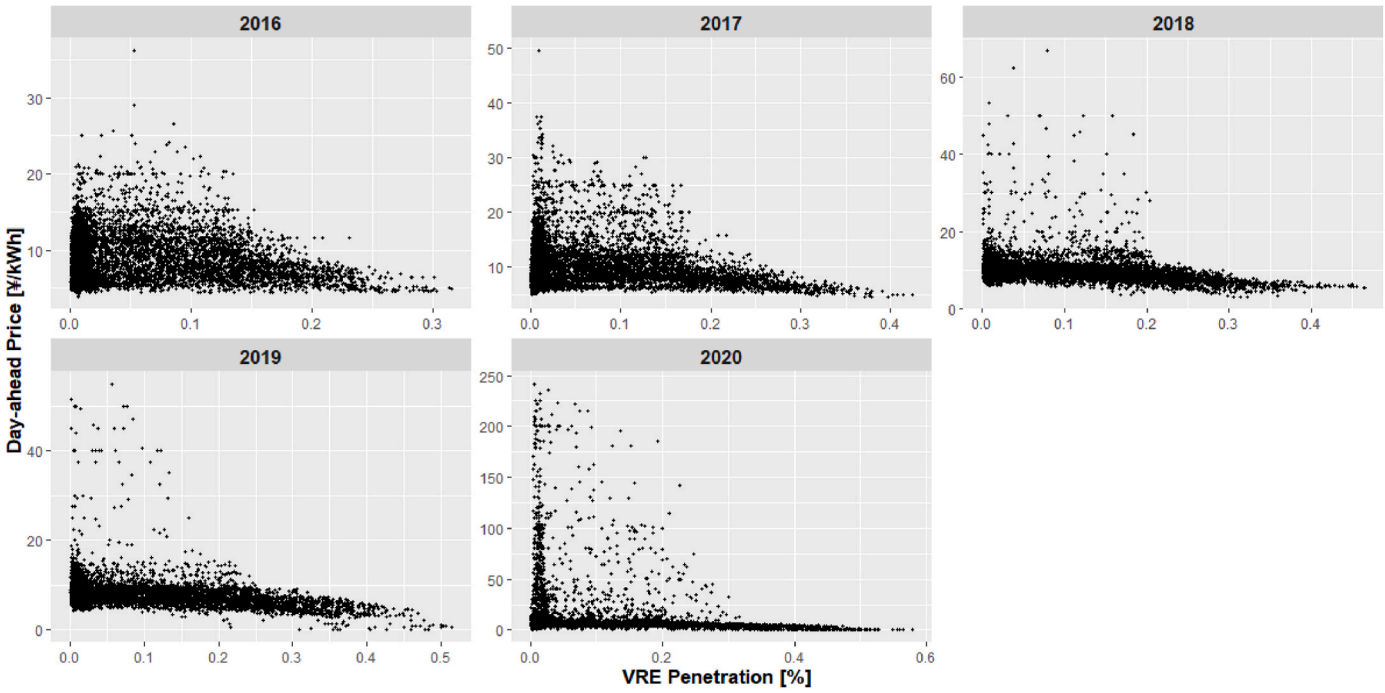


Fig. 11. MCP against hourly national VRE penetration.

Table 3
The MOE for additional GWh of VRE.

Study	Research Country	Period	MOE €/MWh	
			Wind	Solar
Clò et al. (2015)	Italy	2005–2013	4.2	2.3
Zipp (2017)	Germany	2011–2013	1.08–1.54	1.03–1.45
Quint et al., -2019	US	2008–2016	1.07–2.61	
Keeley et al. (2020)	Germany	2010–2017	0.88	0.93
Sakaguchi and Hidemichi (2021)	Japan	FY 2016–FY 2019	2.77	0.68
This Study	Japan	FY2016-FY2020	5.74	0.21

healthy operation of the hydroelectric storage, whereas the charging cycles have upper and lower marginal limits. In addition, although the high penetration of VRE, which overran 50% in 2020, is a significant factor that drives spot prices to nearly zero, it is a major cause of extreme price volatility by leading to higher applications of hydroelectric storage as the costliest solution to balance VRE intermittency. Therefore, the study recommends immediate investment in a real-time balance market to facilitate rapid and economic cross-regional supply and demand balance. Furthermore, commissarial cross-regional virtual power plants are trending methodologies to economically balance spatiotemporal price spikes.

Data statement

The data analyzed in this research is available publicly by the subject authorities. The data sources are mentioned in Table 1, and the cross-bonding information is listed in the references section.

Table 5
Results of regional spot prices interdependency.

	Estimate	Std. Error	t value	Pr(>|t|)
Chubu	0.1150	0.0080	14.288	<2e-16 ***
Chugoku	−2.8500	0.6640	−4.296	1.74e-05 ***
Hokkaido	0.0238	0.0017	13.719	<2e-16 ***
Hokuriku	−3.4300	0.8360	−4.102	4.10e-05 ***
Kansai	6.2700	1.5000	4.184	2.87e-05 ***
Kyushu	0.3600	0.0130	27.771	<2e-16 ***
Shikoku	0.1240	0.0298	4.168	3.08e-05 ***
Tokyo	0.0386	0.0109	3.534	0.00041 ***
Tohoku	0.3480	0.0111	31.249	<2e-16 ***
JCV	0.0087	4.93E-04	17.621	<2e-16 ***

Wald test: 2.617e+05 on 10 and 43813 DF
Residual standard error: 1417 on 43813 degrees of freedom
Multiple R-Squared 0.9835
p-value: <2.2e-16

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

Table 4
Results of correlation analysis.

	Chubu	Chugoku	Hokkaido	Hokuriku	Kansai	Kyushu	Shikoku	Tokyo	Tohoku	MCP
Chubu	1.0000									
Chugoku	0.9905	1.0000								
Hokkaido	0.7447	0.7511	1.0000							
Hokuriku	0.9905	0.9999	0.7509	1.0000						
Kansai	0.9906	0.9999	0.7511	0.9999	1.0000					
Kyushu	0.9799	0.9885	0.7526	0.9884	0.9884	1.0000				
Shikoku	0.9901	0.9996	0.7512	0.9994	0.9995	0.9883	1.0000			
Tokyo	0.8263	0.8336	0.8830	0.8336	0.8337	0.8296	0.8338	1.0000		
Tohoku	0.8288	0.8359	0.8861	0.8359	0.8359	0.8322	0.8361	0.9972	1.0000	
MCP	0.9451	0.9511	0.8387	0.9511	0.9511	0.9486	0.9511	0.9287	0.9312	1.0000

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Table A1
Diagnostic tests of the interdependency econometric

	df1	df2	statistic	p-value
Weak instruments (Kansai_Price)	22	43793	11.21	<2e-16 ***
Weak instruments (Kyushu_Price)	22	43793	551.58	<2e-16 ***
Wu-Hausman	2	43811	51.79	<2e-16 ***
Sargan	20	NA	2864.23	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1.

References

Bobinaitė, V., Juozapavičienė, A., Snieška, V., 2006. Correlation of electricity prices in European wholesale power markets. Eng. Econ. 49 (4). https://www.ceeol.com/search/article-detail?id=161612. (Accessed 28 February 2022).

Bunn, D., Andresen, A., Chen, D., Westgaard, S., 2016. Analysis and forecasting of electricity price risks with quantile factor models. Energy J. 37 (1) https://doi.org/10.5547/01956574.37.1.dbun.

Busu, M., ClodniȚchi, R., MureȘan, M., 2019. A correlation analysis of the spot market prices of the Romanian electricity sector. Management & Marketing. Challenges for the Knowledge Society 14 (1), 150–162. https://doi.org/10.2478/mmcks-2019-0010.

Cech, M., 2016. Panel Regression Analysis of Electricity Prices and Renewable Energy in the European Union. https://mpra.ub.uni-muenchen.de/74640/1/MPRA_paper_74601.pdf. (Accessed 8 May 2022).

Chubu Electric Power Grid. Demand data. https://powergrid.chuden.co.jp/denkiyoho/. (Accessed 28 February 2022).

Chugoku Electric Power Transmission and Distribution Company. Demand data. https://www.energia.co.jp/nw/service/retailer/data/area/index.html. (Accessed 28 February 2022).

Clò, S., Cataldi, A., Zoppoli, P., 2015. The merit-order effect in the Italian power market: the impact of solar and wind generation on national wholesale electricity prices. Energy Pol. 77, 79–88. https://doi.org/10.1016/j.enpol.2014.11.038.

Dickey, D., Fuller, W.A., 1979. Distribution of the estimators for time series regressions with a unit root. J. Am. Stat. Assoc. 74 (366), 427–431. https://doi.org/10.1080/01621459.1979.10482531.

Farzaneh, Hooman, 2019. Energy supply models. In: Farzaneh, Hooman, Energy Systems Modeling: Principles and Applications. Springer Singapore Pte. Limited, pp. 81–105. https://doi.org/10.1007/978-981-13-6221-7.

Grubb, M., Newbery, D., 2018. UK electricity market reform and the energy transition: emerging lessons. Energy J. 39 (6) https://doi.org/10.5547/01956574.39.6.mgru.

Hausman, J.A., 1978. Specification tests in econometrics. Econometrica: J. Econom. Soc. 1251–1271. https://doi.org/10.2307/1913827.

Hokkaido Electric Power Network. Demand data. https://www.hepco.co.jp/network/renewable_energy/fixedprice_purchase/supply_demand_results.html. (Accessed 28 February 2022).

Hokuriku Electric Power Transmission and Distribution Company. Demand data. http://www.rikuden.co.jp/nw_jyukyudata/area_jisseki.html. (Accessed 28 February 2022).

Honda, Y., 1985. Testing the error components model with non-normal disturbances. Rev. Econ. Stud. 52 (4), 681–690. https://doi.org/10.2307/2297739.

Japan Electric Power Exchange (JEPX), 2022. Trading information. http://www.jepx.org/english/market/index.html. (Accessed 28 February 2022).

Japan Electric Power Information Center, 2019. The electric power Industry in Japan 2019. Executive Summary. https://www.jepic.or.jp/en/data/pdf/epijJepic2019.pdf. (Accessed 28 February 2022).

Japan Electric Power Information Center, 2021. The Electric Power Industry in Japan 2021. Executive Summary. https://www.jepic.or.jp/en/data/pdf/epijJepic2021.pdf. (Accessed 28 February 2022).

Jin, L., Chen, C., Wang, X., Yu, J., Long, H., 2020. Research on information disclosure strategies of electricity retailers under new electricity reform in China. Sci. Total Environ. 710, 136382 https://doi.org/10.1016/j.scitotenv.2019.136382.

Kansai Transmission and Distribution. Demand data. Available online at: https://www.kansai-td.co.jp/denkiyoho/area-performance.html. (Accessed 28 February 2021).

Keeley, A.R., Matsumoto, K.I., Tanaka, K., Sugiawan, Y., Managi, S., 2020. The impact of renewable energy generation on the spot market price in Germany: ex-post analysis using boosting method. Energy J. 41 (Special Issue) https://doi.org/10.5547/01956574.41.SI1.akee.

King, M.L., Wu, P.X., 1997. Locally optimal one-sided tests for multiparameter hypotheses. Econom. Rev. 16 (2), 131–156. https://doi.org/10.1080/07474939708800379.

Kyushu Electric Power Transmission and Distribution. Demand data. https://www.kyuden.co.jp/td_service_wheeling_rule-document_disclosure.html. (Accessed 28 February 2021).

Makishi, Sakaguchi, Fujii, Hidemichi, 2021. The impact of variable renewable energy penetration on wholesale electricity prices in Japan between FY 2016 and 2019. Frontiers in Sustainability 2, 2673–4524. https://doi.org/10.3389/frsus.2021.770045.

Ministry of Economy, Trade and Industry, 2013. Report of the Electricity System Reform Expert Subcommittee. https://www.meti.go.jp/english/policy/energy_environment/electricity_system_reform/pdf/201302Report_of_Expert_Subcommittee.pdf. (Accessed 28 February 2022).

Ministry of Economy, Trade and Industry, 2015. Long-term energy supply and demand Outlook. . http://www.meti.go.jp/english/press/2015/pdf/0716_01a.pdf.

Ministry of Finance. Trade statistics of Japan. https://www.customs.go.jp/toukei/info/index_e.htm. (Accessed 28 February 2021).

Morstyn, T., Teytelboym, A., Hepburn, C., McCulloch, M.D., 2019. Integrating P2P energy trading with probabilistic distribution locational marginal pricing. IEEE Trans. Smart Grid 11 (4), 3095–3106. https://doi.org/10.1109/TSG.2019.2963238.

Mosquera-López, S., Nursimulu, A., 2019. Drivers of electricity price dynamics: comparative analysis of spot and futures markets. Energy Pol. 126, 76–87. https://doi.org/10.1016/j.enpol.2018.11.020.

Nepal, R., Jamasb, T., 2012. Market integration, efficiency, and interconnectors: the Irish single electricity market. https://dx.doi.org/10.2139/ssrn.2025628.

Paraschiv, F., Erni, D., Pietsch, R., 2014. The impact of renewable energies on EEX day-ahead electricity prices. Energy Pol. 73, 196–210. https://doi.org/10.1016/j.enpol.2014.05.004.

Quint, D., Dahlke, S., 2019. The impact of wind generation on wholesale electricity market prices in the midcontinent independent system operator energy market: an empirical investigation. Energy 169, 456–466. https://doi.org/10.1016/j.energy.2018.12.028.

Renewable Energy Institute, 2020. Recommendations for power system restructuring toward further deployment of renewables. https://www.renewable-ei.org/en/activities/reports/20201030.php. (Accessed 28 February 2022).

Sensfuß, F., Ragwitz, M., Genoese, M., 2008. The merit-order effect: a detailed analysis of the price effect of renewable electricity generation on spot market prices in Germany. Energy Pol. 36 (8), 3086–3094. https://doi.org/10.1016/j.enpol.2008.03.035.

Shah, D., Chatterjee, S., 2020. A comprehensive review on day-ahead electricity market and important features of world's major electric power exchanges. Int. Transact. Electric. Energy Syst. 30 (7), e12360 https://doi.org/10.1002/2050-7038.12360.

Sakaguchi, Makishi, Hidemichi, Fujii, 2021. The Impact of Variable Renewable Energy Penetration on Wholesale Electricity Prices in Japan Between FY 2016 and 2019. Front. Sustain. 2, 2673–4524. https://doi.org/10.3389/frsus.2021.770045.

Sirin, S.M., Yilmaz, B.N., 2020. Variable renewable energy technologies in the Turkish electricity market: quantile regression analysis of the merit-order effect. Energy Pol. 144, 111660 https://doi.org/10.1016/j.enpol.2020.111660.

TEPCO Power Grid. Demand data. https://www.tepco.co.jp/forecast/html/area_data-j.html. (Accessed 28 February 2022).

Tohoku Electric Power Network. Demand data. https://setsuden.nw.tohoku-epco.co.jp/download.html. (Accessed 28 February 2022).

Urpelainen, J., Yang, J., 2019. Global patterns of power sector reform, 1982–2013. Energy Strategy Rev. 23, 152–162. https://doi.org/10.1016/j.esr.2018.12.001.

Welisch, M., Ortner, A., Resch, G., 2016. Assessment of RES technology market values and the merit-order effect–an econometric multi-country analysis. Energy Environ. 27 (1), 105–121. https://doi.org/10.1177/0958305X16638574.

Wen, L., Suomalainen, K., Sharp, B., Yi, M., Sheng, M.S., 2022. Impact of wind-hydro dynamics on electricity price: a seasonal spatial econometric analysis. Energy 238, 122076. https://doi.org/10.1016/j.energy.2021.122076.

Williams, J.H., Ghanadan, R., 2006. Electricity reform in developing and transition countries: a reappraisal. Energy 31 (6–7), 815–844. https://doi.org/10.1016/j.energy.2005.02.008.

Woo, C.K., Moore, J., Schneiderman, B., Ho, T., Olson, A., Alagappan, L., et al., 2016. Merit-order effects of renewable energy and price divergence in California's day-

ahead and real-time electricity markets. Energy Pol. 92, 299–312. https://doi.org/10.1016/j.enpol.2016.02.023.

Wooldridge, Jeffrey M., 2010. Basic linear unobserved effects panel data models. In: Wooldridge, Jeffrey M. Econometric Analysis of Cross Section and Panel Data. MIT Press, pp. 281–344. https://doi.org/10.1007/s00712-003-0589-6.

Zipp, A., 2017. The marketability of variable renewable energy in liberalized electricity markets–an empirical analysis. Renew. Energy 113, 1111–1121. https://doi.org/10.1016/j.renene.2017.06.072.